



The Bank for International Development

Case Study Analysis

Borrowing Costs in Yen and Dollars:
An Assessment of Currency Selection Strategy

Harvard Business School Case N2-289-034

Gonçalo Marques	100563800@alumnos.uc3m.es
Francisco Cordeiro Batista	100556195@alumnos.uc3m.es
David Onilă	100540705@alumnos.uc3m.es
Sebastian Kopecky	100564869@alumnos.uc3m.es

International Finance — Master in Management

Universidad Carlos III de Madrid (UC3M)

March 15, 2026

Contents

1	Introduction	2
2	Question 1: Comparison of Nominal Borrowing Costs	2
3	Question 2: Interest Rates and Inflation — The Fisher Effect	3
4	Question 3: Effective Borrowing Cost Analysis	4
4.1	Real Interest Rate Comparison	4
4.2	Real Interest Rate Differential	6
4.3	Exchange Rate Risk and PPP Deviations	7
5	Question 4: Currency Preference Criteria	8
6	Question 5: International Parity Conditions	9
6.1	Forward Parity: Forward vs. Spot Exchange Rates	9
6.2	Purchasing Power Parity: Exchange Rates and Inflation	10
6.3	Interest Rate Parity: Interest Rates and Forward Rates	11
6.4	Summary of Parity Condition Performance	12
7	Question 6: Implications for BID's Borrowing Costs	12
7.1	Implications of PPP Deviations	12
7.2	Implications of IRP Deviations	12
7.3	Implications of Forward Parity Deviations	13
7.4	Strategic Implications for BID	13
8	Recommendation	14
8.1	Borrowing Strategy	14
8.2	The October 2015 Issue	14
9	Conclusion	15

1. Introduction

The Bank for International Development (BID) is a AAA-rated supranational institution that raises most of its funds through borrowings in international capital markets. BID hedges all currency positions by matching borrowings in a given currency with loans and low-risk investments in the same currency. This leaves one open question: *which currency* should BID borrow in at any given time?

This report uses 118 monthly observations of 90-day Euro-deposit interest rates, quarterly CPI inflation, and spot and forward exchange rates for the yen and dollar, covering January 2006 to April 2016. All rates are expressed on a continuously compounded 90-day basis, so real interest rates are obtained by subtracting inflation from the nominal rate. Since both the forward rates and the inflation data cover 90-day horizons, the realised spot exchange rate change is measured over three months (i.e., comparing the spot rate at time t with the rate at $t + 3$), yielding 115 usable observations for the parity condition tests.

We address six questions about BID's borrowing strategy: (i) how nominal borrowing costs compare, (ii) the Fisher Effect, (iii) effective borrowing costs through real interest rates, (iv) when BID should prefer one currency over the other, (v) how well international parity conditions held empirically, and (vi) what the deviations from parity imply for BID's costs.

2. Question 1: Comparison of Nominal Borrowing Costs

Compare borrowing costs in yen and dollars over the period. How do the nominal rates in each currency compare?

Over 2006–2016, nominal borrowing costs were consistently lower in Japanese yen than in U.S. dollars. The 90-day Euro-yen rate generally ranged between about 0.1% and 3.6%, while the Euro-dollar rate ran higher, between roughly 1.2% and 4.5% (Figure 1).

The mean quarterly yen nominal rate was 1.54%, compared to 2.49% for the dollar—a persistent spread of approximately 95 basis points. This gap was especially pronounced during the period 2008–2012, when Japanese rates fell close to zero amid the Bank of Japan's accommodative monetary policy, while U.S. rates remained significantly higher. From 2013 onwards, the two rates converged somewhat, as U.S. rates declined and Japanese rates stabilised around 1.5%.

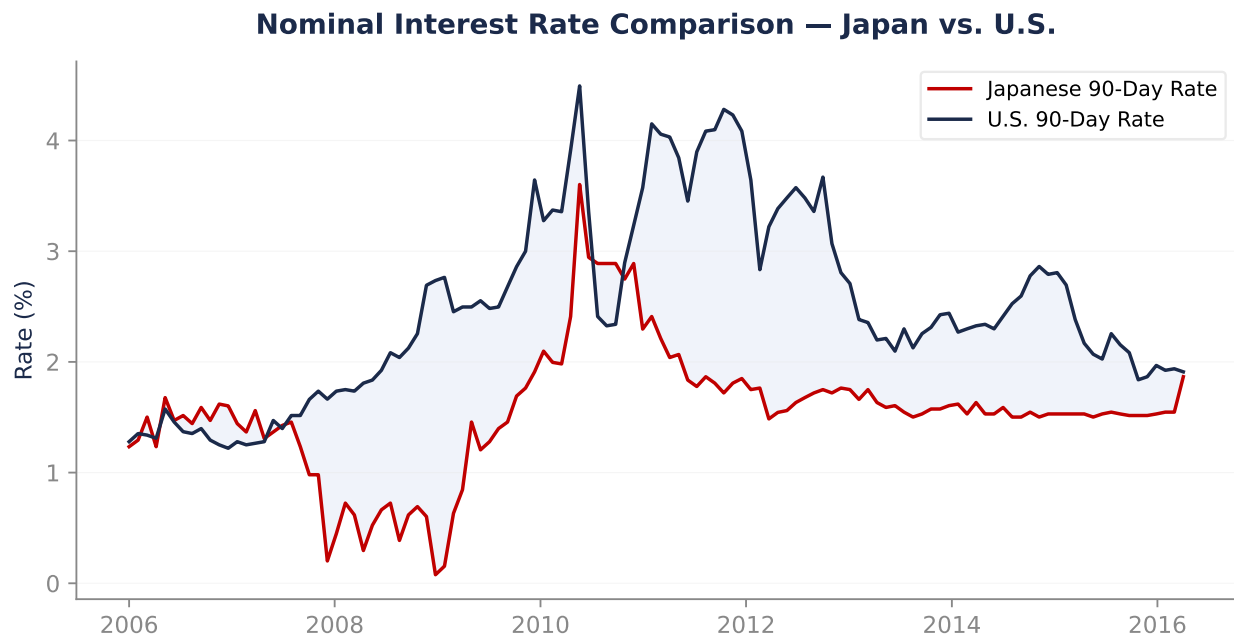


Figure 1: Nominal 90-day interest rates for Japan and the U.S. (2006–2016). Japanese rates were systematically lower throughout the sample period.

However, nominal rates alone do not tell the full story. Exchange rate movements and inflation differentials may offset the apparent advantage of lower yen rates. As the following sections show, the “cheapness” of yen borrowing in nominal terms requires adjustment for purchasing power changes and currency risk.

3. Question 2: Interest Rates and Inflation — The Fisher Effect

What is the relationship between interest rates and inflation rates in each currency?

The **Fisher Effect** posits that nominal interest rates incorporate an inflation premium, such that:

$$i \approx r + \pi^e \quad (1)$$

where i is the nominal interest rate, r is the real interest rate, and π^e is the expected inflation rate. Countries with higher inflation should therefore exhibit higher nominal interest rates, as lenders demand compensation for the reduction of purchasing power.

U.S. inflation was generally higher and more stable than Japanese inflation over the sample (Figure 2). U.S. quarterly CPI inflation typically ranged between 0.5% and 4.2%, averaging 1.71%. Japanese inflation was lower and more volatile, averaging only 1.03%, and frequently dipping below zero—particularly in 2008–2009 and again in 2012–2014.

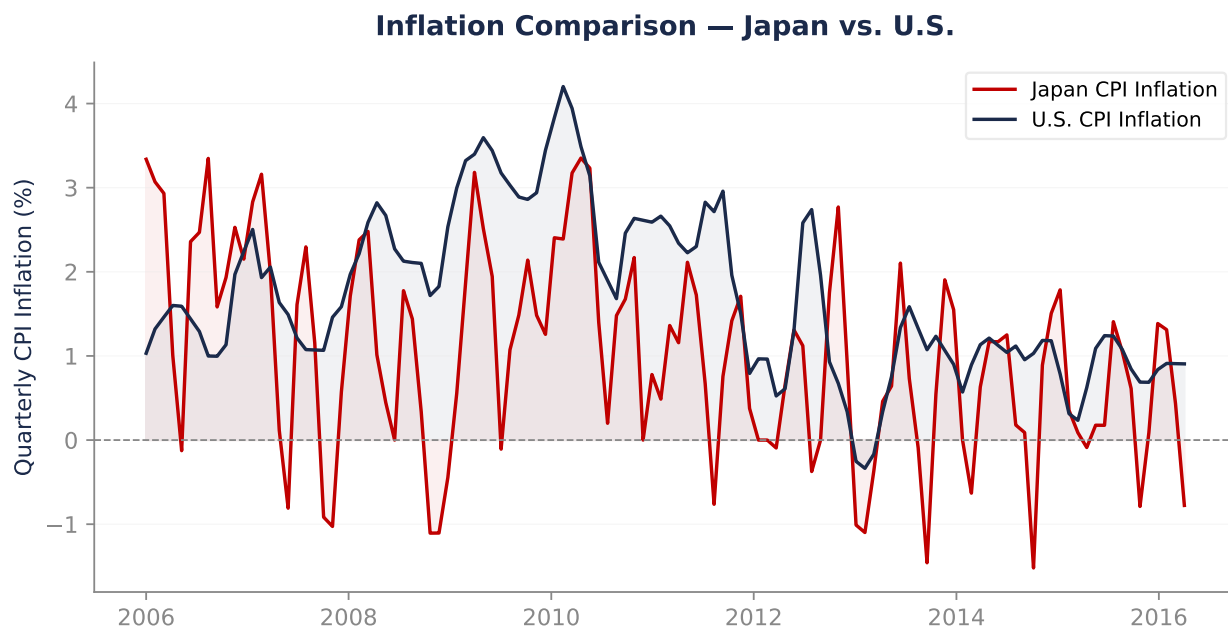


Figure 2: Quarterly CPI inflation rates for Japan and the U.S. (2006–2016). The higher and more stable U.S. inflation is consistent with its higher nominal interest rates.

The pattern is consistent with the Fisher Effect: higher U.S. inflation corresponds to higher U.S. nominal rates, while Japan’s lower (and often negative) inflation corresponds to lower nominal rates. For BID, this means that much of the nominal rate gap between yen and dollar reflects the *inflation differential* rather than a genuine difference in real borrowing costs—which motivates looking at real interest rates directly.

4. Question 3: Effective Borrowing Cost Analysis

How should BID analyse its effective borrowing costs over the period 2006–2015?

The right way for BID to assess borrowing costs is through **real interest rates** and **real interest rate differentials**, not nominal rates, since real rates capture the actual cost of borrowing after adjusting for inflation. Beyond that, BID must also account for exchange rate risk and deviations from international parity conditions, which can substantially change the effective cost of foreign-currency borrowing.

4.1 Real Interest Rate Comparison

Since all rates in the dataset are continuously compounded, the real interest rate is simply:

$$r_{\text{real}} = i_{\text{nominal}} - \pi_{\text{actual}} \quad (2)$$

Figure 3 plots Japanese against U.S. real interest rates. If real rates were equalized across countries (as the international Fisher Effect would predict), the observations would cluster

along the 45-degree line. Instead, there is wide dispersion, indicating persistent deviations from real interest rate parity.

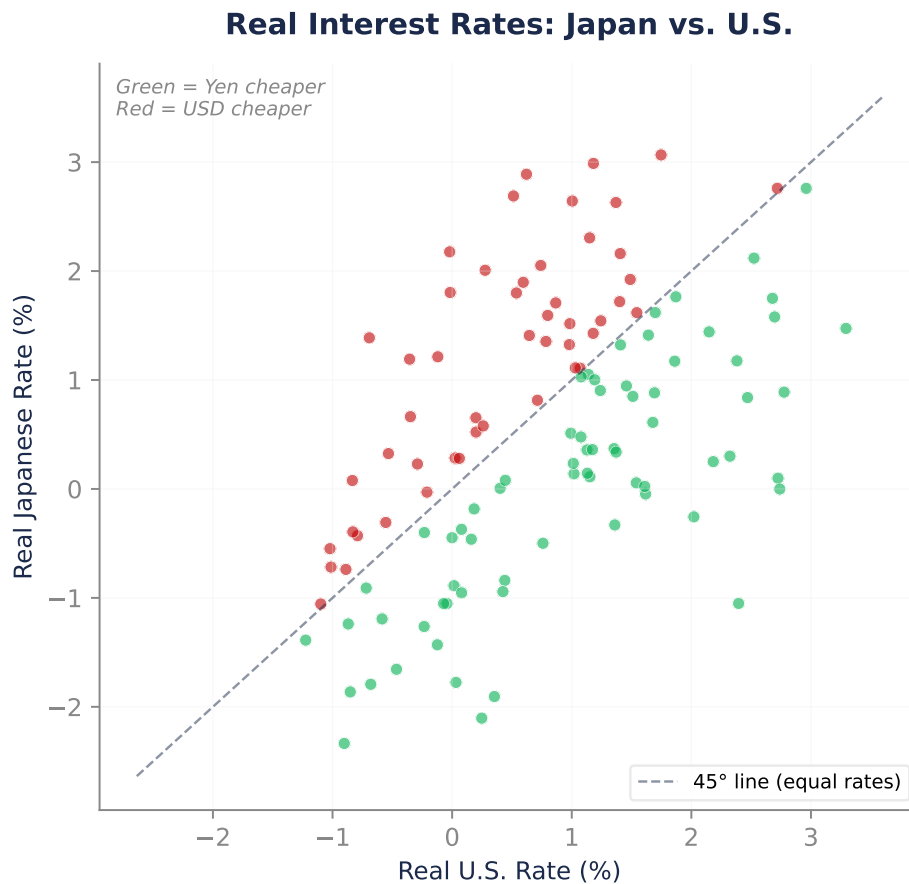


Figure 3: Scatter plot of real interest rates (Japan vs. U.S.). The dispersion around the 45-degree line indicates that real rates were not equalized across the two countries.

Figure 4 traces real rates over time. Japanese real rates were noticeably more volatile than U.S. ones—a consequence of Japan’s erratic inflation combined with relatively stable nominal rates. Both series were generally positive, though Japanese real rates occasionally turned negative during deflationary spells.

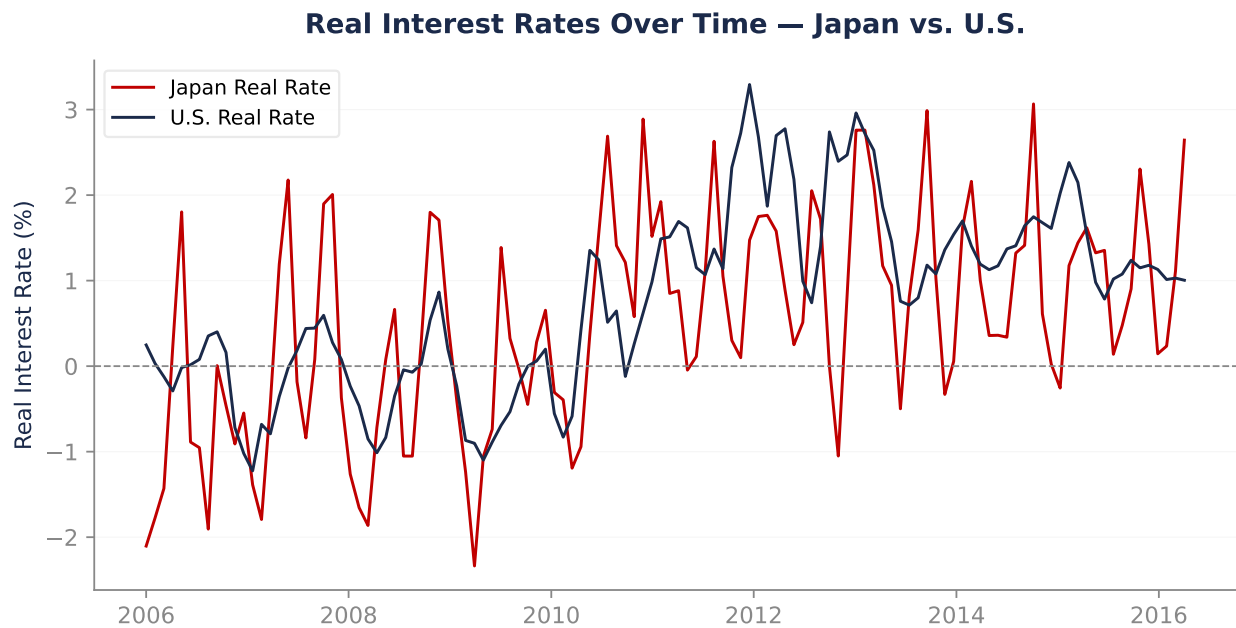


Figure 4: Real interest rates over time for Japan and the U.S. Japanese real rates were more volatile, driven by fluctuations in Japanese inflation.

4.2 Real Interest Rate Differential

The real interest rate differential (Japan minus U.S.) is the most direct measure of relative borrowing cost. Figure 5 shows this differential oscillating around zero throughout the period.

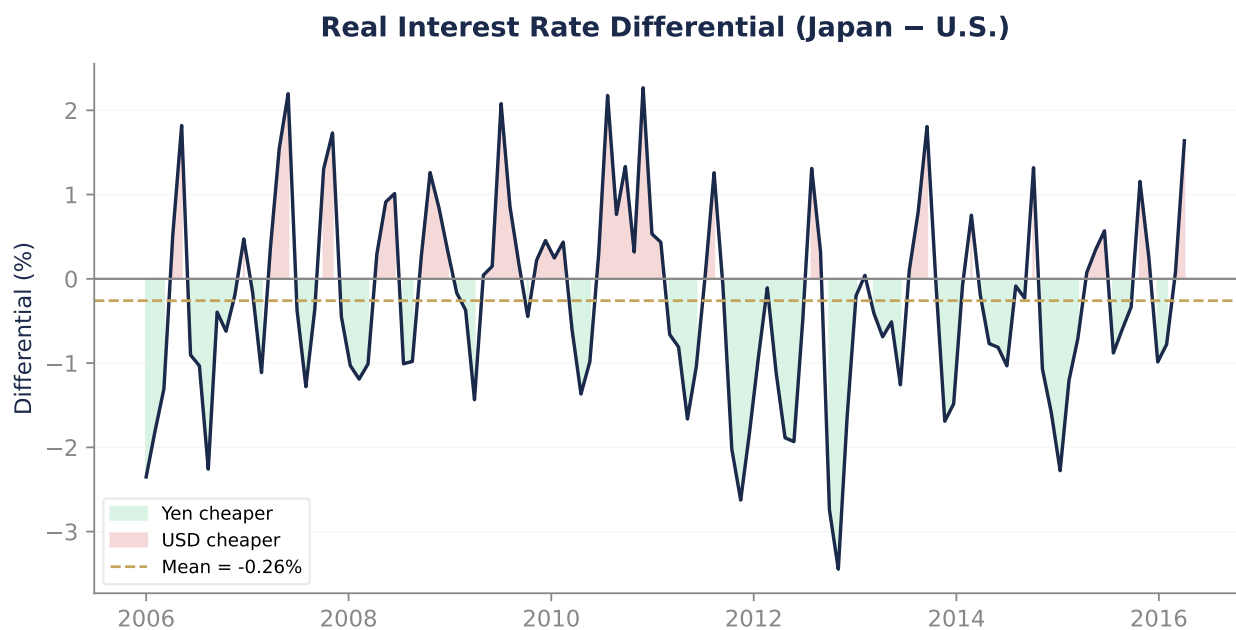


Figure 5: Real interest rate difference (Japan minus U.S.). Negative values indicate that borrowing in yen was cheaper in real terms. The shaded areas highlight the direction of the cost advantage.

Key summary statistics for the real interest rate differential are presented in Table 1.

Table 1: Summary statistics for the real interest rate differential (Japan minus U.S.)

Statistic	Value
Mean	−0.2593%
Standard Deviation	1.1403%
Minimum	−3.4461%
Maximum	+2.0960%
Observations	118
% of Observations < 0 (Yen cheaper)	58.5%

The mean of −0.2593% confirms that yen borrowing was marginally cheaper in real terms over the sample. But the standard deviation of 1.14%—roughly 4.4 times the mean—tells the more important story: the advantage was highly unstable. Yen was cheaper in about 58.5% of months, while the dollar was cheaper in the remaining 41.5%.

4.3 Exchange Rate Risk and PPP Deviations

BID must also consider exchange rate effects. Borrowing in yen exposes BID to the risk that the yen appreciates against the dollar, which would increase the dollar value of repayments. The **Purchasing Power Parity (PPP)** condition predicts that exchange rate changes should offset inflation differentials:

$$\Delta s \approx \pi_{\text{Japan}} - \pi_{\text{US}} \quad (3)$$

Figure 6 shows that PPP deviations were large and persistent, often exceeding $\pm 20\%$ annualized. Exchange rates clearly did not adjust in line with inflation differences, adding considerable uncertainty to any cross-currency borrowing cost comparison.

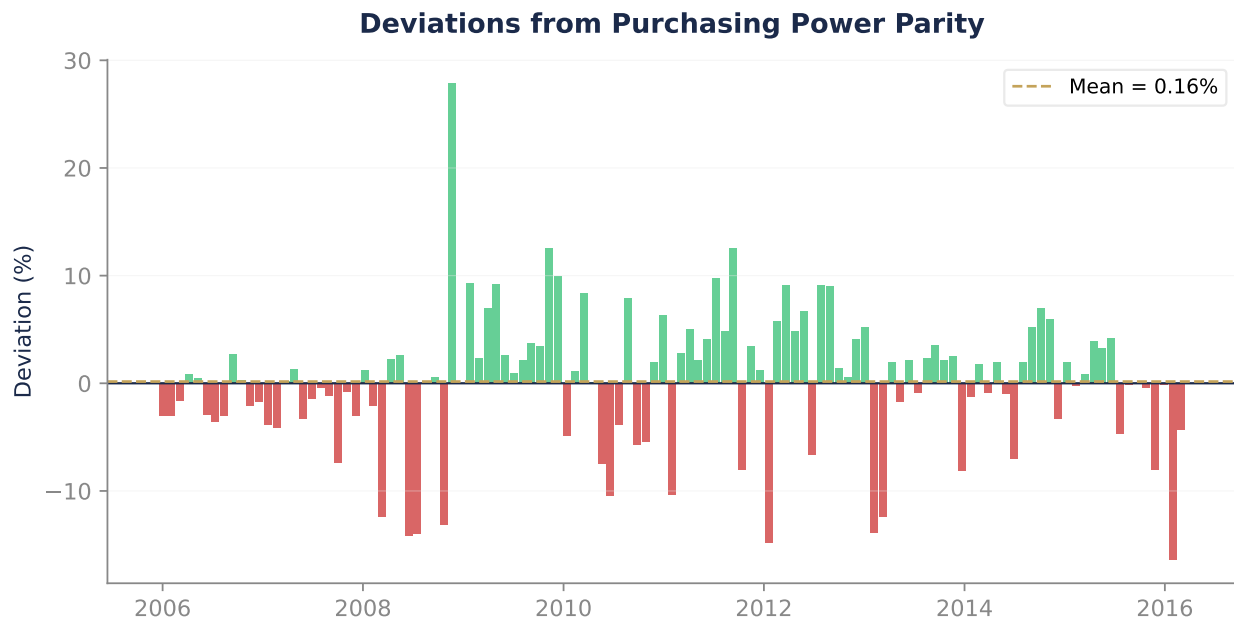


Figure 6: Deviation from Purchasing Power Parity. Large and persistent deviations indicate that exchange rates did not adjust according to inflation differentials.

In short, BID should not stop at comparing nominal rates. Effective borrowing cost analysis requires comparing real interest rates across currencies, tracking the real interest rate differential over time, assessing exchange rate risk through PPP deviations, and evaluating whether forward rates compensate for interest rate differences—the subject of the parity analysis below.

5. Question 4: Currency Preference Criteria

Under what circumstances should BID expect to prefer borrowing in one currency rather than the other?

BID should prefer to borrow in whichever currency offers the lowest **effective real borrowing cost** after accounting for inflation and exchange rate changes.

The starting point is the real interest rate differential. When Japanese real rates are below U.S. real rates (i.e., when the differential in Figure 5 is negative), yen borrowing is cheaper in real terms. During 2006–2016, this was the case about 58.5% of the time, with yen borrowing roughly 26 basis points cheaper on average.

But real rates are only part of the picture. BID must also consider **expected exchange rate movements**. Even if yen rates are lower, yen borrowing becomes less attractive if the yen is expected to appreciate significantly, since appreciation raises the dollar-equivalent repayment cost and can more than offset the interest rate advantage. As Figure 6 showed, exchange rate movements were large and hard to predict—the yen appreciated from roughly

¥204/\$ in 2006 to ¥112/\$ by early 2016, a cumulative move of about 45%.

If **Interest Rate Parity** holds and forward contracts are available, BID should compare *hedged* borrowing costs. Under perfect IRP, the forward rate exactly offsets the interest rate differential, making hedged costs identical across currencies. As Section 6 will show, IRP did not hold perfectly, so residual differences in hedged costs did persist.

Conversely, BID may prefer dollars when U.S. real rates are lower or when the yen is expected to appreciate sharply—roughly 41.5% of months in our sample.

The upshot is that a static preference for one currency is not optimal. The relative cost advantage shifts over time, creating timing opportunities that are potentially valuable but difficult to exploit reliably.

6. Question 5: International Parity Conditions

Consider the various international parity conditions. During 2006–2015, how well did these conditions describe the observed relationships?

We evaluate three international parity conditions—Forward Parity, Purchasing Power Parity, and Interest Rate Parity—by measuring how closely each held over the sample period.

6.1 Forward Parity: Forward vs. Spot Exchange Rates

Forward Parity states that the forward exchange rate should be an unbiased predictor of the future spot exchange rate:

$$F_t = E[S_{t+1}] \implies \Delta s_{t+1} \approx f_t - s_t \quad (4)$$

where $f_t - s_t$ is the forward premium/discount. Since the forward contracts are 90-day instruments and the data is monthly, the realised spot change is computed over a three-month horizon: $\Delta s = \ln(S_{t+3}) - \ln(S_t)$. The deviation from Forward Parity is:

$$\text{Dev}_{FP} = \Delta s_{t+3} - (f_t - s_t) \quad (5)$$

Forward Parity performed poorly in the short run (Figure 7). Deviations were large and volatile, with a standard deviation of 11.74% and a mean of +0.50%. The mean is close to zero, suggesting no strong systematic bias, but the variability is enormous—individual quarterly deviations frequently exceeded $\pm 10\%$ and sometimes $\pm 20\%$.

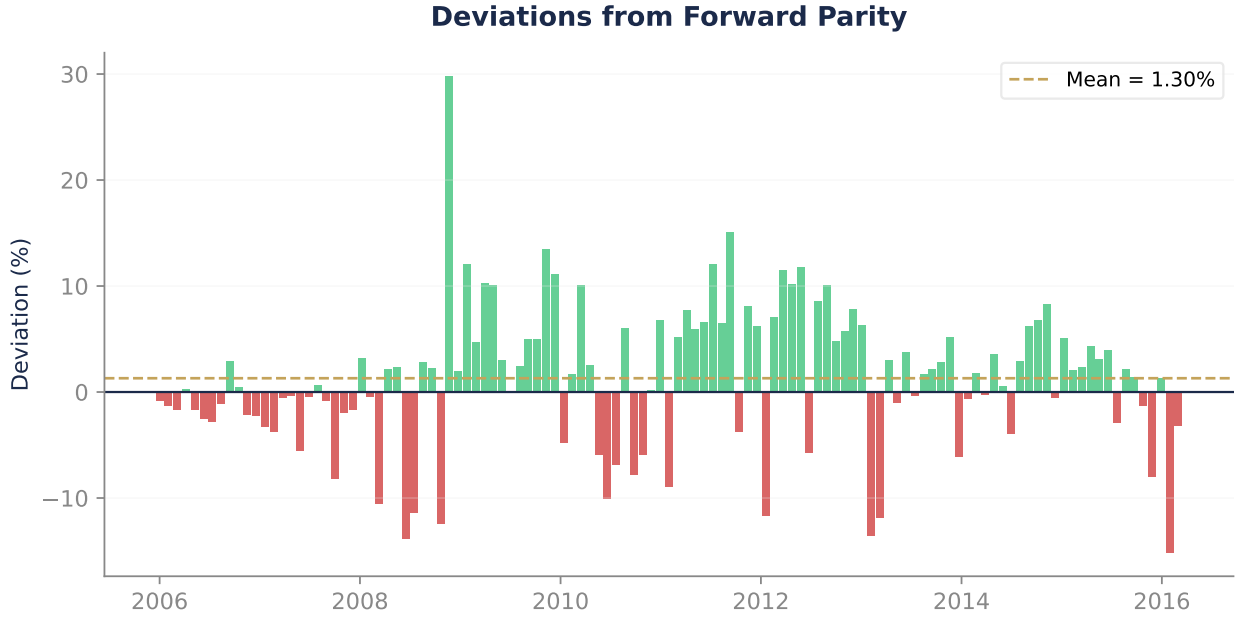


Figure 7: Deviation from Forward Parity. The large deviations indicate that forward rates were poor short-run predictors of future spot rates.

That said, the deviations did fluctuate around zero, suggesting that forward rates were at least *unbiased* predictors over the long run—even though their accuracy in any single quarter was limited.

6.2 Purchasing Power Parity: Exchange Rates and Inflation

Purchasing Power Parity (PPP) states that changes in the exchange rate should reflect differences in national inflation rates:

$$\Delta s \approx \pi_{\text{Japan}} - \pi_{\text{US}} \quad (6)$$

As with Forward Parity, the spot change is measured over a three-month horizon to match the quarterly inflation data: $\Delta s = \ln(S_{t+3}) - \ln(S_t)$.

PPP performed poorly over 2006–2016 (Figure 6). The deviations were large, with a standard deviation of 11.33% and occasional excursions beyond $\pm 20\%$. The mean deviation was -0.64% , which is economically small relative to the volatility.

This is consistent with a well-known result in international finance: PPP holds reasonably well over long horizons (5–10 years or more) but breaks down in the short run. Real exchange rate movements driven by capital flows, terms-of-trade shocks, and speculation overwhelm the gradual inflation-driven adjustment that PPP predicts.

6.3 Interest Rate Parity: Interest Rates and Forward Rates

Covered Interest Rate Parity (CIP) states that the interest rate differential between two currencies should equal the forward premium/discount:

$$i_{\text{Yen}} - i_{\text{USD}} = f_t - s_t \quad (7)$$

The deviation from IRP is:

$$\text{Dev}_{\text{IRP}} = (i_{\text{Yen}} - i_{\text{USD}}) - (f_t - s_t) \quad (8)$$

Figure 8 confirms that IRP held much better than either PPP or Forward Parity. The standard deviation of IRP deviations was only 0.93%, roughly twelve times smaller than for PPP. That said, the mean deviation of +0.85% points to a small but systematic departure from perfect parity.

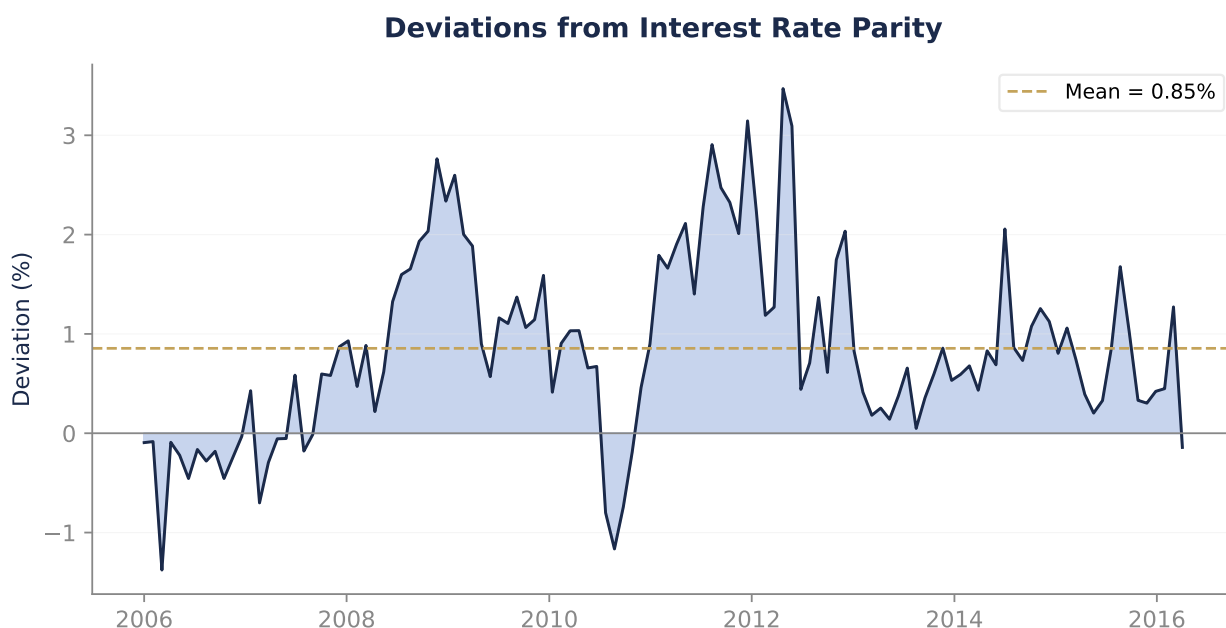


Figure 8: Deviation from Interest Rate Parity (Nominal Rate Difference minus Forward Premium). IRP held better than PPP or Forward Parity, but small systematic deviations persisted.

This is expected. Covered interest arbitrage ensures that large IRP deviations are quickly exploited and eliminated by market participants. The small residual deviations likely reflect transaction costs, capital controls (particularly in Japan during parts of the sample), and credit risk differences between Euro-deposits and sovereign instruments.

6.4 Summary of Parity Condition Performance

Table 2: Summary of deviations from international parity conditions

Parity Condition	Mean Dev. (%)	Std Dev. (%)	Assessment
Interest Rate Parity (IRP)	+0.85	0.93	Holds
Forward Parity (FP)	+0.50	11.74	Fails
Purchasing Power Parity (PPP)	−0.64	11.33	Fails

7. Question 6: Implications for BID’s Borrowing Costs

How does the performance of the parity conditions affect the effective borrowing costs faced by BID?

The deviations from the three parity conditions feed directly into BID’s borrowing decisions. Each deviation channel affects the bank’s effective costs in a different way.

7.1 Implications of PPP Deviations

The poor short-term performance of PPP is the most consequential finding for BID. If PPP held perfectly, exchange rate movements would exactly offset inflation differentials, and real borrowing costs would be equal across currencies regardless of which nominal rate BID chose. In practice, the large and persistent deviations (standard deviation of 11.33%) mean several things.

First, exchange rate risk is real and substantial. Borrowing in yen looks cheap in nominal terms, but unexpected yen appreciation can sharply increase dollar-equivalent repayment costs. The yen’s cumulative move from ¥204/\$ to ¥112/\$ over the sample period illustrates how large this effect can be.

Second, real borrowing cost differences persist. Because exchange rates do not adjust to equalize real rates, the observed mean real interest rate differential of −0.26% in favour of yen is a genuine (if small) cost saving, not an artifact of expected exchange rate changes.

Third, currency selection matters. If PPP held, BID would be indifferent across currencies and the treasurer’s market-timing strategy would be pointless. The failure of PPP means that currency choice does affect effective costs, which is precisely what makes the internal debate worth having.

7.2 Implications of IRP Deviations

IRP held well (standard deviation of only 0.93%), and this has practical consequences.

Because IRP approximately holds, hedging with forward contracts largely eliminates borrowing cost differences across currencies. Under perfect IRP, a hedged yen borrowing and a

dollar borrowing cost exactly the same, since the forward premium/discount exactly offsets the interest rate gap.

There are small residual opportunities—the mean IRP deviation of +0.85% and its variability suggest that BID may occasionally find slightly cheaper hedged borrowing in one currency. But these savings are small relative to the PPP-driven risks discussed above and may well be eaten up by transaction costs.

The corollary is that the treasurer’s unhedged speculation is risky. If IRP holds, the interest rate differential is already “priced in” by forward markets. The only way for the treasurer to beat the market is by making unhedged bets on exchange rate movements—exactly the area where PPP fails and forecasting is hardest.

7.3 Implications of Forward Parity Deviations

The poor performance of Forward Parity (standard deviation of 11.74%) cuts both ways.

On one hand, forward rates are clearly not reliable short-run predictors. The large deviations between actual spot changes and forward predictions mean BID cannot rely on forward rates to forecast whether the yen will appreciate or depreciate.

On the other hand, the deviations confirm that, *ex post*, there were large timing opportunities. In some quarters the yen appreciated far more than the forward rate implied (making yen borrowing expensive), while in others it depreciated (making yen borrowing cheap). But exploiting these opportunities *ex ante* requires forecasting exchange rate movements, which is notoriously difficult.

The mean deviation was only +0.50%, close to zero, suggesting no strong systematic “forward discount bias” over this period. The forward rate was roughly unbiased on average, but its accuracy in any individual quarter was very poor.

7.4 Strategic Implications for BID

Putting the three parity results together, BID faces a clear strategic trade-off.

The *conservative path* is to hedge all currency exposure using forward contracts. Since IRP approximately holds, hedged borrowing costs are roughly the same in yen and dollars, and BID avoids the exchange rate risk exposed by PPP deviations. Under this approach, the choice of borrowing currency barely matters, and the treasurer’s market-timing efforts add little value.

The *speculative path*—actively choosing currencies based on the treasurer’s exchange rate views—can in principle reduce costs, because PPP deviations create real cost differences. But the treasurer would need to consistently forecast exchange rate movements better than the market, and the Forward Parity analysis shows just how volatile exchange rate surprises are (standard deviation of nearly 12%). Even occasional forecasting successes may be offset

by large errors.

8. Recommendation

8.1 Borrowing Strategy

BID should move toward a rules-based approach to currency selection, setting internal allocation targets and hedging the bulk of its non-dollar exposure with forward contracts. The case for this is straightforward: IRP holds well enough that hedged borrowing costs are nearly identical across currencies, so hedging removes the exchange rate risk that PPP deviations create—at very little cost. For a AAA-rated institution whose credibility depends on conservative financial management, taking on avoidable currency risk for marginal savings is hard to justify.

That does not mean eliminating all discretion. PPP failures confirm that real borrowing cost differences between currencies are genuine, and the standard deviation of the real rate differential (1.14%) means that the gap can be meaningful in any given quarter. A rigid allocation with no flexibility would forgo these opportunities entirely. The treasurer should retain the ability to leave a modest share of borrowing unhedged—no more than 15–20% of a given issue—but only when the real rate differential is clearly in favour of one currency *and* the forward premium is unusually large by historical standards.

The default should always be to hedge. The burden of proof should fall on the case for leaving exposure open, not the other way around.

8.2 The October 2015 Issue

For the \$500 million issue, BID should **issue in U.S. dollars** at 7.95% rather than yen at 7.75%.

The yen’s 20 basis point nominal advantage is small—and it looks even smaller against the data. The yen appreciated roughly 45% over the preceding decade (from ¥204/\$ to ¥112/\$), and quarterly exchange rate surprises routinely reached 11–12%. A 20 bp saving is negligible next to that kind of volatility. The forward rate (¥211.61/\$) sits marginally above the spot (¥211.50/\$), so the market is pricing almost no expected movement—but the historical record shows how little that means in practice.

A dollar issue also has practical advantages. Since a large share of BID’s assets are dollar-denominated, there is no need to hedge, which saves on transaction costs. And with 33% of BID’s debt already in dollars versus only 14% in yen (Exhibit 1), a dollar issue fits more naturally into the existing currency mix.

If BID preferred yen for diversification reasons, it should fully hedge the exposure via forwards. Given that IRP approximately holds, the effective cost would end up close to the

dollar rate anyway—which rather proves the point that the 20 bp nominal gap is not a real saving.

9. Conclusion

This analysis of BID’s borrowing costs in yen and dollars from 2006 to 2016 points to several conclusions.

Nominal rates were consistently lower in yen, with an average spread of about 95 basis points—consistent with Japan’s lower inflation and the Fisher Effect. After adjusting for inflation, yen borrowing was still marginally cheaper, with a mean real rate differential of -0.26% in favour of yen. But this advantage was neither large nor stable: the standard deviation of the differential (1.14%) was more than four times the mean, and in over 40% of months the dollar was actually cheaper.

Of the three parity conditions, Interest Rate Parity held reasonably well, meaning that hedged borrowing costs were approximately equalized across currencies. PPP and Forward Parity, by contrast, performed poorly, confirming that exchange rate risk is both substantial and hard to forecast.

We side with the critics who called for a rules-based approach. The treasurer’s market-timing strategy is, at bottom, a bet on exchange rate forecasting—and the data show just how poorly that works over short horizons. For a AAA-rated institution, the downside risk of getting it wrong outweighs the marginal savings of getting it right. BID should set allocation targets, hedge most of its exposure, and save tactical flexibility for situations where the cost advantage is clearly large enough to warrant the risk. For October 2015, that means issuing in dollars.